

APOORV DIXIT

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PROJECTS

- **LLM Inference Optimization** - Built an LLM inference benchmarking framework for GPT-2 (124M), comparing PyTorch FP32/FP16, **SDPA**, torch.compile, **ONNX** Runtime, and **TensorRT** across batch sizes 1–16 and sequence lengths 64–256; measured latency, throughput, and GPU memory, achieving **~3.4× higher throughput** (110K vs 32K tokens/s) and **~46% lower latency** (16.5 ms vs 30.3 ms) with TensorRT over ONNX.
- **LLM Serving Infrastructure** - Built a production-style GPU inference serving platform benchmarking HuggingFace **FastAPI**, **vLLM**, **SGLang**, **TensorRT-LLM**, and NVIDIA **Triton** for GPT-2 FP16 workloads on an RTX 4080 Laptop GPU; measured p50/p95 latency, throughput, and GPU utilization, with SGLang achieving **3.2× lower latency** (80 vs 257 ms) and **3.2× higher throughput** (12.4 vs 3.9 req/s) than HuggingFace and Triton baselines while TensorRT-LLM delivered comparable performance through optimized LLM inference.
- **GPU Profiling & Analysis** - Profiled GPT-2 inference on an RTX 4080 Laptop GPU using PyTorch **Profiler**, CUDA Events, and NVIDIA **Nsight** Systems; analyzed batch sizes 1–32, identified GEMM-dominated execution (~76% kernel activity), synchronization overhead, and throughput saturation beyond batch size 8, achieving peak throughput of ~105K tokens/sec
- **LLM Quantization Benchmarking** - Built a GPU inference benchmarking framework to evaluate FP32, FP16, INT8, and INT4 precision modes for GPT-2 on an NVIDIA RTX 4080; achieved 3.2× latency reduction, 3× throughput improvement (120K tokens/s), and 48% memory savings with FP16 while quantifying the accuracy and performance impact of low-bit quantization through perplexity-based evaluation
- **AI Workload Analysis** - Conducted comparative performance analysis of **GPT-2**, **DistilGPT-2**, **BERT-base**, and **ResNet-50** on an RTX 4080 GPU, measuring latency, throughput, memory footprint, FLOPs, and Model FLOP Utilization (MFU); identified batching saturation at batch sizes 8–16 and achieved up to 39% MFU during FP16 inference
- **CUDA Kernel Optimization** - Developed custom **CUDA kernels** (vector addition, tiled matrix multiplication, and warp-shuffle reduction) as PyTorch C++ extensions; achieved 630 GB/s memory bandwidth, 486 GFLOPS compute throughput, and up to 41× speedup over CPU baselines on an RTX 4080 Laptop GPU

EXPERIENCE

JPMorgan Chase & Co.

Jun 2025 - Present | Jersey City, NJ

Quantitative Analytics Associate

- Architected and productionized a GPT-5 (**AzureOpenAI**) + Ensemble **RAG** system (**LangChain**, Chroma Vector DB, BM25, Parent Document Retriever) to auto-generate structured risk review drafts from 80-400 page model documents, reducing enterprise model review cycle time by **33%** (3 months to 2 months) across **500+** analysts.
- Implemented a **TabTransformer** in PyTorch on 300K loan records (11.5:1 imbalance), benchmarking against **Logistic Regression** (0.58 ROC-AUC) and **XGBoost** (0.65), achieving 0.61 ROC-AUC while evaluating attention-based feature interaction learning for credit risk modeling
- Conducted assessment of a **Regression-based** Auto loans recovery forecasting model (20K records) supporting an \$85B auto-loans portfolio, executing marginal vs conditional **R²** analysis, **KS/CSI** stability tests, **VIF** multicollinearity checks, **heteroskedasticity** diagnostics, and backtesting to surface structural risks prior to CCAR-aligned regulatory review
- Reconstructed and audited an **XGBoost** regression pipeline (20K datapoints) predicting 5 venture finance metrics using log transforms, k-fold CV, Random Search hyperparameter tuning, Recursive Feature Elimination, and SHAP explainability, identifying 7 risks before publication

JPMorgan Chase & Co.

Jun 2024 - Aug 2024 | Jersey City, NJ

Quantitative Analyst Intern

- Built an LLM-based benchmarking platform evaluating 8 document extraction models, achieving 94% F1 on CoNLL-2003
- Developed a **OpenCV** based document stress-testing engine generating 10+ noisy variants per file to evaluate robustness of 8 extraction models

Siemens Digital Industries Software

Jul 2021 - Jul 2023 | Pune, India

Associate Software Engineer

- Designed a two-stage AI pipeline (**OpenCV + Tesseract/Google Vision OCR + SpaCy/BERT NER**) to convert scanned PDF manufacturing drawings into SolidEdge-native 2D CAD files by extracting geometric primitives (73% shape accuracy) and metadata tables (90% NER accuracy), integrating outputs via SolidEdge APIs to prototype a **top customer-requested** automation feature in the CAD modeling domain
- Developed an early **GPT-3.5-powered conversational assistant** integrated into a web-based CAD editor (Zel-X) that leveraged help documentation to guide users through complex modeling workflows, prototyping an LLM-driven support interface to improve user experience
- Delivered 9 features and resolved 50+ issues across two cloud SaaS products while increasing automated test coverage to 98%.

SKILLS

Python, C++, CUDA, PyTorch, Hugging Face Transformers, ONNX Runtime, TensorRT, vLLM, NVIDIA Triton Inference Server, TensorRT-LLM, SGLang, torch.compile, NVIDIA Nsight Systems, PyTorch Profiler, Azure OpenAI, LangChain, ChromaDB, BM25, RAG, FastAPI, XGBoost, TabTransformer, SHAP, Statsmodels, OpenCV, Tesseract OCR, Google Vision API, PyMuPDF, Pandas, AsyncIO

EDUCATION

University of Southern California

Aug 2023 - May 2025 | Los Angeles, USA

Master of Science in Computer Science

GPA - 3.73 / 4

Pune Institute of Computer Technology

Aug 2017 - April 2021 | Pune, India

Bachelor of Engineering in Computer Engineering

GPA - 9.14 / 10